

Validation of `RMitemRestscoreCutoff()`

Nested null simulation of the bootstrap item-restscore test

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Purpose

`restscore_asymptotic_null.qmd` showed that the asymptotic item-restscore test in `iarm::item_restscore()` is miscalibrated under a true Rasch model. With BH over the items, the chance of at least one false flag ranged from 30.5 percent (20 dichotomous items, $n = 150$, mistargeted by 1.5 logits) down to about 1 percent (9 polytomous items, $n \geq 1000$). Underfit was flagged too rarely in every cell.

`RMitemRestscoreCutoff()` replaces the asymptotic reference distribution with a parametric bootstrap that refits the model in every iteration. Studentising the observed minus expected difference by the bootstrap mean and SD should remove both causes identified in the null study, the small-sample bias and the correlation between observed and expected gamma. This study tests that claim in the cells where the asymptotic test did worst, and compares the two data-generating processes (`dgp = "resample"` and `"conditional"`).

Design

Each simulated dataset is generated under a true Rasch model, exactly as in the null study, and then analysed the way a user would:

1. `RMitemRestscoreCutoff(d, iterations = 400, dgp = ...)`, the package default number of iterations.
2. `RMitemRestscore(d, cutoff = co)` with `correction = "fwer"` (the default), `"fdr_bh"` and `"none"`.
3. `RMitemRestscore(d)` without a cutoff, the asymptotic BH rule, on the same dataset for a paired comparison.

Cells. Chosen from the null study to span its worst liberal and most conservative results.

cell	items	μ	n	asymptotic BH family-wise rate
<code>dich_mis_150</code>	20 dichotomous	1.5	150	30.5%

cell	items	μ	n	asymptotic BH family-wise rate
dich_tgt_300	20 dichotomous	0	300	6.2%
poly_tgt_150	9 × 4 categories	0	150	4.2%
poly_tgt_1000	9 × 4 categories	0	1000	0.9%

Arms. Both DGPs in every cell, on the same datasets.

Replications. 500 datasets per cell. The Monte Carlo SE of a 5 percent family-wise rate is then 0.97 points, so rates between 3.1 and 6.9 percent are consistent with nominal.

Outcomes.

- Family-wise rate: at least one item flagged, for WY (FWER), BH and the asymptotic rule.
- Per-item marginal rejection at $\alpha = .05$, split by direction (overfit and underfit, nominal 2.5 percent each). This tests whether the asymmetry of the asymptotic test is gone.
- The distribution of marginal p-values, which should be uniform.

Cost. About 0.1 s per bootstrap iteration, so 40 s per dataset and arm. 4 cells × 2 arms × 500 datasets is about 45 CPU-hours, or 4.5 hours on 10 cores. Results are saved per cell and arm, so an interrupted render resumes where it stopped. Set `R_DATASETS` lower for a quick run.

```
suppressMessages({
  devtools::load_all("...", quiet = TRUE)
  library(parallel)
  library(dplyr)
  library(tidyr)
  library(ggplot2)
})
options(rgl.useNULL = TRUE)

R_DATASETS <- 500
B           <- 400
THETA_SD    <- 1.5
NCORES      <- 10
RES_DIR     <- "restscore_cutoff_validation"
dir.create(RES_DIR, showWarnings = FALSE)
```

Cells and data generation

The item sets and the generator are the same as in the null study.

```
item_sets <- list(
  dich = as.list(seq(-2, 2, length.out = 20)),
  poly = lapply(seq(-1.5, 1.5, length.out = 9), function(l) l + c(-1.2, 0, 1.2))
)

cells <- data.frame(
  cell = c("dich_mis_150", "dich_tgt_300", "poly_tgt_150", "poly_tgt_1000"),
  set  = c("dich", "dich", "poly", "poly"),
  mu   = c(1.5, 0, 0, 0),
  n    = c(150, 300, 150, 1000),
  stringsAsFactors = FALSE
)
arms <- c("resample", "conditional")
cells
```

cell	set	mu	n
dich_mis_150	dich	1.5	150
dich_tgt_300	dich	0.0	300
poly_tgt_150	poly	0.0	150
poly_tgt_1000	poly	0.0	1000

```
sim_pcm <- function(theta, thr) {
  n <- length(theta)
  sapply(thr, function(d) {
    m <- length(d)
    eta <- outer(theta, 0:m) -
      matrix(c(0, cumsum(d)), n, m + 1, byrow = TRUE)
    p <- exp(eta - apply(eta, 1, max))
    p <- p / rowSums(p)
    cp <- p %*% upper.tri(diag(m + 1), diag = TRUE)
    rowSums(runif(n) > cp)
  })
}
```

One dataset

A dataset with an unobserved category is regenerated rather than discarded, since the question is how the bootstrap behaves on data the package can fit. The number of redraws is recorded.

```
one_dataset <- function(rep, cell_row, dgp) {
  thr <- item_sets[[cell_row$set]]
  seed <- 1e6 * match(cell_row$cell, cells$cell) + rep
  set.seed(seed)
  redraws <- 0L
  repeat {
    d <- as.data.frame(sim_pcm(rnorm(cell_row$n, cell_row$mu, THETA_SD), thr))
    names(d) <- paste0("I", seq_along(thr))
    full <- all(mapply(function(x, m) all(tabulate(x + 1, m + 1) > 0),
                      d, lengths(thr)))
    if (full) break
    redraws <- redraws + 1L
  }

  out <- tryCatch({
    co <- suppressMessages(RMitemRestscoreCutoff(
      d, iterations = B, parallel = FALSE, seed = seed,
      cutoff_method = "quantile", dgp = dgp
    ))
    run <- function(correction) {
      suppressMessages(RMitemRestscore(
        d, cutoff = co, correction = correction, output = "dataframe"
      ))
    }
    fw <- run("fwer")
    bh <- run("fdr_bh")
    nc <- run("none")
    asy <- suppressMessages(RMitemRestscore(d, output = "dataframe"))
  })
}
```

```

data.frame(
  rep = rep,
  item = fw$Item,
  difference = fw$Difference,
  p = fw$p_restscore,
  flag_marginal = nc$Flagged,
  padj_fwer = fw$padj_restscore,
  padj_bh = bh$padj_restscore,
  flag_asy = asy$Flagged,
  actual_iterations = co$actual_iterations,
  redrows = redrows,
  stringsAsFactors = FALSE
)
}, error = function(e) NULL)
out
}

```

Run

One file per cell and arm. Delete a file to re-run that part.

```

timing <- list()
for (i in seq_len(nrow(cells))) {
  for (dgp in arms) {
    f <- file.path(RES_DIR, paste0(cells$cell[i], "_", dgp, ".rds"))
    if (file.exists(f)) next
    t0 <- Sys.time()
    res <- mclapply(seq_len(R_DATASETS),
                   function(r) one_dataset(r, cells[i, ], dgp),
                   mc.cores = NCORES, mc.preschedule = FALSE)
    saveRDS(list(res = res,
                 elapsed_min = as.numeric(Sys.time() - t0, units = "mins")),
            f)
  }
}

long <- do.call(rbind, lapply(seq_len(nrow(cells)), function(i) {
  do.call(rbind, lapply(arms, function(dgp) {
    f <- file.path(RES_DIR, paste0(cells$cell[i], "_", dgp, ".rds"))
    x <- readRDS(f)
    ok <- Filter(Negate(is.null), x$res)
    if (length(ok) == 0) return(NULL)
    cbind(cell = cells$cell[i], dgp = dgp, do.call(rbind, ok),
          elapsed_min = x$elapsed_min, failed = length(x$res) - length(ok))
  })))
}))
long$cell <- factor(long$cell, levels = cells$cell)

long |>
  group_by(cell, dgp) |>
  summarise(datasets = n_distinct(rep),
            failed = first(failed),
            redrows = sum(redrows[!duplicated(rep)]),
            min_iterations = min(actual_iterations),

```

```
minutes = round(first(elapsed_min), 1),
.groups = "drop")
```

cell	dgp	datasets	failed	redraws	min_iterations	minutes
dich_mis_150	conditional	500	0	0	344	25.7
dich_mis_150	resample	500	0	0	363	18.9
dich_tgt_300	conditional	500	0	0	400	32.8
dich_tgt_300	resample	500	0	0	400	20.4
poly_tgt_150	conditional	500	0	0	258	26.2
poly_tgt_150	resample	499	1	0	280	21.4
poly_tgt_1000	conditional	500	0	0	400	47.7
poly_tgt_1000	resample	500	0	0	400	33.8

Results

Family-wise rate

Proportion of datasets with at least one flagged item, nominal 5 percent. The asymptotic column is the same in both arms, since it does not use the bootstrap.

```
fw <- long |>
  group_by(cell, dgp, rep) |>
  summarise(wy = any(padj_fwer < .05),
            bh = any(padj_bh < .05),
            asymptotic = any(flag_asy != ""),
            .groups = "drop") |>
  group_by(cell, dgp) |>
  summarise(datasets = n(),
            across(c(wy, bh, asymptotic), \(x) round(100 * mean(x), 1)),
            .groups = "drop")
fw
```

cell	dgp	datasets	wy	bh	asymptotic
dich_mis_150	conditional	500	4.2	2.8	29.6
dich_mis_150	resample	500	5.4	3.6	29.6
dich_tgt_300	conditional	500	6.8	6.2	10.0
dich_tgt_300	resample	500	9.2	8.2	10.0
poly_tgt_150	conditional	500	3.4	2.8	1.6
poly_tgt_150	resample	499	3.8	4.4	1.6
poly_tgt_1000	conditional	500	5.6	5.0	0.8
poly_tgt_1000	resample	500	7.0	5.6	0.8

Per-item rejection by direction

Marginal bootstrap p-value below .05 (`correction = "none"`), split by the direction label, against the same split for the asymptotic test with BH. The bootstrap should give 2.5 percent in each direction.

```
long |>
  group_by(cell, dgp) |>
  summarise(boot_over = 100 * mean(flag_marginal == "overfit"),
            boot_under = 100 * mean(flag_marginal == "underfit"),
            asy_over = 100 * mean(flag_asy == "overfit"),
            asy_under = 100 * mean(flag_asy == "underfit"),
            .groups = "drop") |>
  mutate(across(where(is.numeric), \(x) round(x, 2)))
```

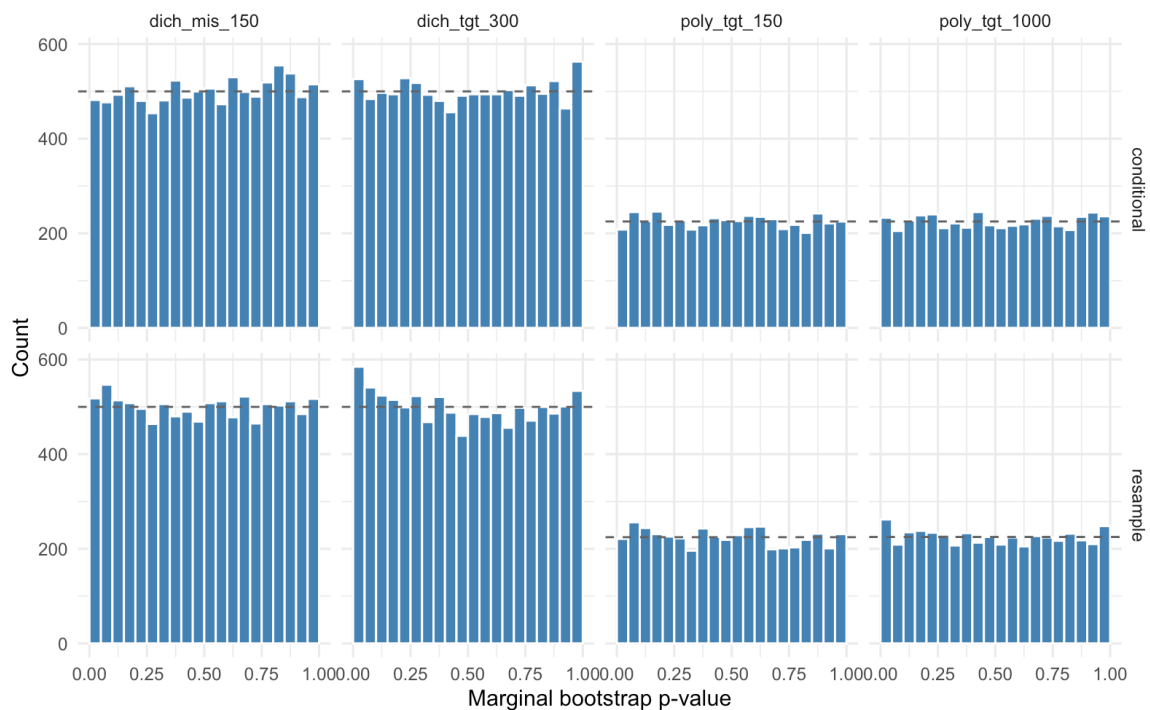
cell	dgp	boot_over	boot_under	asy_over	asy_under
dich_mis_150	conditional	1.77	3.04	1.87	0.03
dich_mis_150	resample	1.91	3.25	1.87	0.03
dich_tgt_300	conditional	2.21	3.05	0.54	0.03
dich_tgt_300	resample	2.49	3.36	0.54	0.03
poly_tgt_150	conditional	1.98	2.64	0.18	0.00
poly_tgt_150	resample	2.14	2.72	0.18	0.00
poly_tgt_1000	conditional	2.40	2.78	0.07	0.02
poly_tgt_1000	resample	2.69	3.13	0.07	0.02

The asymptotic columns are BH-adjusted and so not directly comparable in level with the marginal bootstrap columns. What matters is the ratio of the two tails within each method.

Marginal p-values

Under the null the marginal p-values should be uniform. With $B = 400$ the smallest attainable value is $1/401$, so the histogram is coarse at the left edge.

```
long |>
  ggplot(aes(p)) +
  geom_histogram(breaks = seq(0, 1, 0.05), fill = "steelblue",
                 colour = "white") +
  geom_hline(aes(yintercept = n / 20),
             data = long |> count(cell, dgp),
             linetype = "dashed", colour = "grey40") +
  facet_grid(dgp ~ cell) +
  labs(x = "Marginal bootstrap p-value", y = "Count") +
  theme_minimal()
```



Additional checks

Paired DGP comparison. Both arms analyse the same datasets, so the difference between them is a paired comparison. The table counts datasets flagged by WY under one DGP and not the other.

```
fw_rep <- long |>
  group_by(cell, dgp, rep) |>
  summarise(wy = any(padj_fwer < .05), bh = any(padj_bh < .05),
            asymptotic = any(flag_asy != ""), .groups = "drop")
fw_rep |>
  select(cell, dgp, rep, wy) |>
  pivot_wider(names_from = dgp, values_from = wy) |>
  filter(!is.na(resample), !is.na(conditional)) |>
  group_by(cell) |>
  summarise(resample_only = sum(resample & !conditional),
            conditional_only = sum(!resample & conditional),
            both = sum(resample & conditional), .groups = "drop")
```

cell	resample_only	conditional_only	both
dich_mis_150	11	5	16
dich_tgt_300	18	6	28
poly_tgt_150	4	2	15
poly_tgt_1000	10	3	25

```
fw_rep |>
  group_by(dgp) |>
  summarise(datasets = n(), wy = round(100 * mean(wy), 2),
            bh = round(100 * mean(bh), 2), .groups = "drop")
```

dgp	datasets	wy	bh
conditional	2000	5.00	4.20
resample	1999	6.35	5.45

The `dich_tgt_300` draw. The asymptotic rate in this cell is 10.0 percent against 6.2 percent over 4000 datasets in the null study. The package path and a direct `iarm::item_restscore()` call agree on every one of the 500 datasets, and the null study's seeds give 6.4 percent through the package path (checked outside this document), so the method is the same and the 500 datasets are a high draw. Whether that carries over to the bootstrap arms shows in the WY rate split by the asymptotic decision on the same dataset.

```
fw_rep |>
  group_by(cell, dgp, asymptotic) |>
  summarise(datasets = n(), wy = round(100 * mean(wy), 1),
            .groups = "drop") |>
  pivot_wider(names_from = asymptotic, values_from = c(datasets, wy),
            names_glue = "{.value}_asy_{asymptotic}")
```

cell	dgp	datasets_asy_FALSE	datasets_asy_TRUE	wy_asy_FALSE	wy_asy_TRUE
dich_mis_150	conditional	352	148	3.1	6.8
dich_mis_150	resample	352	148	3.7	9.5
dich_tgt_300	conditional	450	50	4.2	30.0
dich_tgt_300	resample	450	50	6.7	32.0
poly_tgt_150	conditional	492	8	2.8	37.5
poly_tgt_150	resample	491	8	3.1	50.0
poly_tgt_1000	conditional	496	4	4.8	100.0
poly_tgt_1000	resample	496	4	6.2	100.0

Skew of the null. Under the null the observed difference across the 500 datasets is a draw from the same distribution the bootstrap estimates, so its per-item skewness shows the shape the two-sided $|t|$ test folds. The second number is the correlation over items between skewness and the underfit-minus-overfit marginal rejection rate.

```
skew <- function(x) { x <- x - mean(x); mean(x^3) / mean(x^2)^1.5 }
sk <- long |>
  filter(dgp == "conditional") |>
  group_by(cell, item) |>
  summarise(skew = skew(difference),
            asym = 100 * (mean(flag_marginal == "underfit") -
                          mean(flag_marginal == "overfit")),
            .groups = "drop")
sk |>
  group_by(cell) |>
  summarise(mean_skew = round(mean(skew), 3),
            mean_asym = round(mean(asym), 2), .groups = "drop")
```

cell	mean_skew	mean_asym
dich_mis_150	-0.283	1.27
dich_tgt_300	-0.145	0.84
poly_tgt_150	-0.092	0.67
poly_tgt_1000	-0.051	0.38

```
round(cor(sk$skew, sk$asym), 2)
```

```
[1] -0.79
```

Iterations kept. WY rate by whether all 400 iterations succeeded.

```
long |>
  group_by(dgp, rep, cell) |>
  summarise(wy = any(padj_fwer < .05),
            full = first(actual_iterations) == B, .groups = "drop") |>
  group_by(dgp, full) |>
  summarise(datasets = n(), wy = round(100 * mean(wy), 2), .groups = "drop")
```

dgp	full	datasets	wy
conditional	FALSE	103	4.85
conditional	TRUE	1897	5.01
resample	FALSE	211	5.21
resample	TRUE	1788	6.49

Interpretation

1. The bootstrap removes the miscalibration of the asymptotic test. In the worst cell of the null study, 20 dichotomous items mistargeted by 1.5 logits at $n = 150$, the asymptotic BH rule flagged at least one item in 29.6 percent of datasets. On the same datasets the bootstrap gave 4.2 percent (WY) and 2.8 percent (BH) with the conditional DGP. In the most conservative cell, 9 polytomous items at $n = 1000$, the asymptotic rate of 0.8 percent rose to 5.6 percent (WY). The marginal p-values are close to uniform in every cell and arm, and the underfit tail, which the asymptotic test almost never rejected (0.00 to 0.03 percent per item), now rejects 2.6 to 3.4 percent.

2. The conditional DGP is calibrated and the resample DGP is somewhat liberal. Pooled over the four cells (2000 datasets per arm), the WY family-wise rate was 5.0 percent with the conditional DGP and 6.35 percent with resample, and BH gave 4.2 and 5.45 percent. Resample was higher in all four cells, and the paired comparison leaves little room for chance: 43 datasets were flagged by WY under resample only and 16 under conditional only. The marginal rejection rates show the same pattern (resample 4.9 to 5.9 percent, conditional 4.6 to 5.3). The mechanism has not been measured. A likely candidate is that resampled WLE locations do not reproduce the score distribution the observed gamma is conditioned on, which the conditional DGP holds fixed by construction.

3. dich_tgt_300 is a high draw, not a calibration failure. The 500 datasets in this cell gave the asymptotic test 10.0 percent against a long-run 6.2 percent, and datasets the asymptotic test flagged were also flagged by the bootstrap WY in 30 percent of cases, against 4.2 percent for the rest. The bootstrap rates in this cell (6.8 conditional, 9.2 resample) are therefore inflated by the same draw. Reweighting the two groups to the long-run asymptotic rate gives roughly 5.8 and 8.3 percent, a rough figure that leaves the conditional arm within Monte Carlo error and the resample arm still above it.

4. The two directions do not get equal shares of α . The marginal test rejects at about the nominal total, but the underfit tail takes more than half of it (2.6 to 3.4 percent against 1.8 to 2.7 for overfit). The null distribution of the difference is left-skewed, since gamma is bounded at 1, and a two-sided test on $|t|$ splits α evenly only under symmetry. The asymmetry follows the skewness across cells (largest for mistargeted dichotomous items, smallest for polytomous items at $n = 1000$) and across items (correlation -0.79). An equal-tailed p-value would remove it. That change touches `.bootstrap_pvalues()`, which the other cutoff functions share, and is not decided.

5. Failed iterations did not affect the rate. Some datasets kept fewer than 400 iterations (down to 258 in `poly_tgt_150`), because simulated datasets with an unobserved category are rejected. The WY rate was not higher in datasets with fewer iterations.

Implications. The bootstrap test should replace the asymptotic test as the recommended path, and the conditional DGP is the better default for `RMitemRestscoreCutoff()`, at 1.2 to 1.6 times the run time of resample. The study used 400 iterations throughout, so it supports 400 as adequate but says nothing about fewer.